

FORECASTING COVID-19 WITH TEMPORAL HIERARCHIES

A Thesis Presented

by

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ABSTRACT

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Infectious disease forecasting efforts underwent rapid growth during the COVID-19 pandemic, providing guidance for pandemic response and about potential future trends. Yet despite their importance, these short-term forecasting models often struggled to produce accurate real-time predictions of a complex and rapidly-changing system. This gap in accuracy, which persists three years into the pandemic, warrants the exploration and testing of new methods to glean fresh insights.

In this work, we examine the application of the temporal hierarchical forecasting (THieF) methodology to probabilistic forecasting of COVID-19 hospital admissions in the United States. THieF is a recent, innovative forecasting technique that aggregates time-series data into a hierarchy made up of different temporal scales, produces forecasts at each level of the hierarchy, then reconciles those forecasts using optimized weighted forecast combination. While THieF's unique approach has shown substantial accuracy improvements in a diverse range of applications, such as opera-

tions management and predicting emergency room admissions, this technique has not previously been applied to outbreak forecasting.

We generate candidate models formulated using the THieF methodology, which differ by their hierarchy schemes and data transformations, and weighted ensembles of the THieF models. The models are validated using weighted interval score (WIS), and the top-performing subset are compared to two categories of models: simple ARIMA and SARIMA models and operational incident hospitalization models from the US COVID-19 Forecasts Hub. The latter group includes benchmark models like a naive baseline and a median ensemble of all submitted models. Preliminary work demonstrates accuracy improvements in WIS, MAE, and prediction interval coverage from the THieF models over the baseline and simple ARIMA and SARIMA models during the training phase, suggesting potentially similar accuracy gains in the testing phase.

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